

Phani Pavan Kambhampati

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ABOUT ME

A Final-year master's research scholar heading the algorithmic research and applications of neuromorphic computing at IIIT Bangalore. Experience researching spiking neural networks for neuromorphic computing, reinforcement learning based automation and embedded system. Diligent and collaborative researcher with multiple peer-reviewed publications. Looking to secure a doctoral scholar position in the field of neuromorphic computing and spiking neural networks.

EDUCATION

Master of Science by Research in VLSI and Embedded Systems 2023-2025

IIIT Bangalore, India

- ✓ CGPA: 3.9/4.0
- ✓ *Specialization*: Spiking Neural Networks for Neuromorphic Computing
- ✓ *Related coursework*: Optimizations, Reinforcement Learning, Machine Learning, Embedded System Design

Bachelor of Technology in Electronics and Communication 2018-2022

Manipal Institute of Technology, Manipal, India

- ✓ CGPA: 3.1/4.0
- ✓ *Specialization*: Machine Learning for IoT Systems
- ✓ *Thesis*: RUL Estimation of Battery using Basic and Advanced Machine Learning Techniques.

RESEARCH EXPERIENCE

Research Assistant Aug 2022 – Dec 2022

IIIT Bangalore, India

- ✓ Worked on diverse projects including Spiking Neural Networks, Machine Learning, Internet of Things and Embedded Systems.
- ✓ Contributed as a core member of the AutoGrow Team for electronics and software automation, supported patent and research publications alongside supervising in-terns.

Data Analyst Intern Apr 2022 – July 2022

Reliance Jio, Bangalore, India

- ✓ Corrected sensor drift of battery management system by upto 50 cycles by developing an end-to-end pipeline model.
- ✓ Fine-tuned the model using Jio's closed-source 300M data and openly available datasets to generalize the prediction across multiple battery models.

Research Assistant Apr 2021 – Dec 2021

Manipal Institute of Technology, Manipal, India

- ✓ Collaboratively published an object detection dataset, primarily assisting with asset management, frame annotation, model validation and automation scripting.
- ✓ Gathered nearly 10 hours of aerial footage of objects perceived as small in various scenarios and processed them into multiple formats for better usability.

RESEARCH PROJECTS

Closed-loop Automation System with Spiking Networks

- **Skills:** SNN Torch, Physics Simulation, Reinforcement Learning
- ✓ Proposed and evaluated an end-to-end framework for heterogeneous reinforcement learning-based automation system for muscle rehabilitation.
- ✓ Optimized spike timing to achieve over 60% gain in power and latency compared to base SNNs, while retaining 95% of control behaviour.

Adapting Spiking Networks for Regressive Tasks

- **Skills:** SNN Torch, Signal Processing, Physics Simulation
- ✓ Developed novel spike decoders using SNN Torch to adapt spiking networks for regressive open-loop control of a rehabilitation system.
- ✓ Achieved superior prediction performance against SOTA with over 10x estimated power and latency gains.

Custom Neurons for Spiking Network based Classifier

- **Skills:** SNN Torch, Genetic Algorithms, Bio-Signal Acquisition
- ✓ Designed custom brain inspired spiking neurons to assist classifying bio-signals, along with custom architecture search algorithm.
- ✓ Proposed various time-variant and image-based models with 90% to 98% accuracy and significant efficiency gains, tested across several low-power devices.

Green House Automation With Internet of Things

- **Skills:** Gymnasium, Reinforcement Learning, HAL Programming
- ✓ Created and patented IoT and RL-based irrigation system to optimise nutrient delivery to a greenhouse, in collaboration with the Institute of Horticulture Research, Bangalore.
- ✓ Constructed a proof-of-concept system with hardware integration and data warehousing for developing the RL system.

Small Object Detection Dataset Using UAV Video Footage

- **Skills:** Python, CVAT, Video Processing
- ✓ Published a well cited dataset with over 13K images for small object detection and benchmarked it against several aerial datasets like VisDrone and Okutama.
- ✓ Labelled 150K objects in video frames of different scenarios using tools like CVAT and EVA, and trained object detection algorithms including YoloV5, CornerNET, various R-CNNs etc.

PUBLICATIONS

Journal Articles

- ✓ **Phani Pavan K;** Gautam C.; P, J. & Rao, M. (Submitted to IOP Neuromorphic Comp. & Eng.) NeuRehab: A Reinforcement Learning and Spiking Neural Network-Based Rehab Automation Framework. *ArXiv Pre-print*. [arXiv:2512.17841](https://arxiv.org/abs/2512.17841)
- ✓ S, A. B., **Phani Pavan K**, & Rao, M. (2024). MC-QDSNN: Quantized deep evolutionary SNN with Multi-Dendritic Compartment neurons for stress detection using physiological signals. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 44(4): 1313–1325. [10.1109/tcad.2024.3484353](https://doi.org/10.1109/tcad.2024.3484353)
- ✓ Akshatha, K. R., **Phani Pavan K**, Karunakar, A. K., & Shenoy, B. S. (2024). Enhancing person detection in UAV color images: Empirical study and Novel Techniques for improved performance. *IEEE Access*, 12: 140699–140714. [10.1109/access.2024.3467372](https://doi.org/10.1109/access.2024.3467372)
- ✓ KR, A., AK, K., B, S. S., **Phani Pavan K**, Dhareshwar, C. V., & Johnson, D. G. (2022). Manipal-UAV person detection dataset: A step towards benchmarking dataset and algorithms for small object detection. *ISPRS Journal of Photogrammetry and Remote Sensing*, 195: 77–89. [10.1016/j.isprsjprs.2022.11.008](https://doi.org/10.1016/j.isprsjprs.2022.11.008)

Conference Proceedings

- ✓ **Phani Pavan K**, Thoppur, A., R, N., & Rao, M. (2025) Spike Decoders for Regressing Arm Angle for Autonomous Stroke Rehabilitation System. *Proceedings of the IEEE International Symposium on Circuits and Systems (ISCAS), UK*. [10.1109/ISCAS56072.2025.11043614](#)
- ✓ P, J., **Phani Pavan K**, & Rao, M. (2025) AI-Enhanced Motor Power Analysis for Joint Kinematics Prediction and Fault Detection in Rehabilitation Systems. *Proceedings of the Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*. [10.1109/EMBC58623.2025.11253851](#)
- ✓ Kestur, R., **Phani Pavan K**, Rao, M., & GM, P. P. (2025) AutoGrow RL - Adaptive input control of an IoT based Autonomous Greenhouse System for Precision Farming using Reinforcement Learning. *Proceedings of the International Symposium on Industrial Electronics, Canada*. [10.1109/ISIE62713.2025.11124615](#)
- ✓ **Phani Pavan K**, S, A. B., & Rao, M. (2024). Energy Efficient Multi-Modal Stress Detection System with Dynamic Adaptive Spiking Neurons. *Proceedings of the Great Lakes Symposium on VLSI, USA*, 138-143. [10.1145/3649476.3658723](#)
- ✓ S, A. B., **Phani Pavan K**, & Rao, M. (2024). DESPINE: NAS generated Deep Evolutionary Adaptive Spiking Network for Low Power Edge Computing Applications. *Proceedings of the IEEE International Symposium on Quality Electronic Design (ISQED), USA*. [10.1109/isqed60706.2024.10528734](#)
- ✓ S, A. B., Rao, M., & **Phani Pavan K** (2024). Neuromorphic Energy Efficient Stress Detection System using Spiking Neural Network. *Proceedings of the IEEE International Symposium on Circuits and Systems (ISCAS), Singapore*. [10.1109/iscas58744.2024.10558058](#)

PRESENTATIONS

- ✓ Spike Decoders for Regressing Arm Angle for Autonomous Stroke Rehabilitation System, *ISCAS 2025, London, UK*. (27th May, 2025). Oral.
- ✓ Spike Decoders for Regressing Arm Angle for Autonomous Stroke Rehabilitation System, *RISE Open House, IIIT Bangalore, 2025*. Campus Talk.
- ✓ Murmur: A Secure and Low-Energy Audio Communication for the Internet of Bodies, *COMSNETS 2025, Bangalore, India*. (8th Jan, 2025). Oral.
- ✓ Energy Efficient Multi-Modal Stress Detection System with Dynamic Adaptive Spiking Neurons, *GLSVLSI 2024, Florida, USA*. (12th June, 2024). Oral.
- ✓ Neuromorphic Energy Efficient Stress Detection System using Spiking Neural Network, *RISE Open House, IIIT Bangalore, 2024*. Campus Talk.

SKILLS

Technical

- **Programming/Scripting**: Python, Rust, Dart, Bash, Arduino-C
- **Spiking Simulators**: SNN Torch, Spiking Jelly, Sinabs, NIR
- **Machine Learning**: PyTorch, SNN Torch, NumPY, Pandas
- **Reinforcement Learning**: Gymnasium, Mujoco, VREP, Blender, PyGame
- **Other Frameworks**: WebSockets, Flutter, Harmony
- **Markup**: $\text{\LaTeX}$, Markdown

Non-Technical

- **Languages**: Telugu [Native], English [Advanced, IELTS:8], German [Beginner]
- **Hobbies**: Photography, Open-source Contribution, Reading Novels

REFERENCES

- ✓ **Prof. Madhav Rao**, mr@iiitb.ac.in
Head of the Dept. of Electronics and Communication, IIIT Bangalore
- ✓ **Prof. Ramesh Kestur**, rameshkestur@iiitb.ac.in
Adjunct Faculty, IIIT Bangalore, India.
- ✓ **Prof. Kurian Polachan**, kurian.polachan@iiitb.ac.in
Assistant Professor, CDWELab, IIIT Bangalore, India.
- ✓ **Prof. Raghuram Bharadwaj**, raghuram.bharadwaj@iiitb.ac.in
Assistant Professor, SCL Lab, IIIT Bangalore, India.